Latent Factor Models

CP4285 Modern Recommendation Systems — Week 02

CP4285 Instruction Team

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01 Announcements and Context



WEEK 01 RECAP

The User–Item Matrix

R	i_1	i_2	i_3	i_4	i_5
u_1	5	?	2	4	?
u_2	?	4	?	1	3
u_3	?	?	?	?	?
u_4	3	?	4	?	2
u_5	?	2	?	5	?
u_6	5	4	5	2	4
u_7	5	1	5	1	5
u_8	?	?	4	?	?

$$R \in \mathbb{R}^{|U| \times |I|}$$

- $u \in U$: user; $i \in I$: item
- r_{ui}: observed feedback
- Ω: observed (u, i) pairs

> ⚠ **Warning**
>
> A ? is usually **unobserved**, not a negative rating.

> 🤔 **To think about**
>
> What user types might be represented in the matrix?
> What other forms can "ratings" take?

From a Score to a Useful List

Prediction

Estimate a value such as $\hat{r}_{ui}$ for one user–item pair.

Question: "Will this user rate this film highly?"

Top-*k* ranking

Order candidate items and present a short list.

Question: "Which five films should appear first, now?"

Accurate scores do not automatically make a useful list.

> 🤔 **To think about**
>
> Does top-*k* apply to users as well as items?

Similarity I: Pearson Correlation

$$\text{Pearson}(u, v) = \frac{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)(r_{vk} - \mu_v)}{\sqrt{\sum_{k \in I_u \cap I_v} (r_{uk} - \mu_u)^2} \sqrt{\sum_{k \in I_u \cap I_v} (r_{vk} - \mu_v)^2}}$$

Overlap: only mutually rated items contribute.

Centred Rating: rating minus that user's mean, so generous and harsh raters are put on a comparable baseline.

Normalisation: turns aligned variation into a bounded correlation-like score.

Use it when

Rating-scale differences matter.

Watch for

Very small overlap can create an unstable correlation.

From Neighbours to a Prediction

$$\hat{r}_{uj} = \mu_u + \frac{\sum_{v \in P_u(j)} \text{sim}(u, v)\,(r_{vj} - \mu_v)}{\sum_{v \in P_u(j)} |\text{sim}(u, v)|}$$

Neighbourhood $P_u(j)$: the top-k similar users who rated item j.

Similarity weight: closer neighbours contribute more.

Baseline and deviation: predict a deviation, then add the target user's own mean back.

Pipeline

1. choose a similarity
2. select valid top-k neighbours
3. aggregate their evidence
4. rank candidate items by $\hat{r}_{uj}$

Pre-Flight Survey · Metadata Summary

PRE-FLIGHT SURVEY · 26 SUBMISSIONS — cohort signals

COMPUTER SCIENCES
85%
22 of 26

BALANCE / PRACTICE
81%
lecture preference

IR KNOWLEDGE
15%
4 of 26

EMBEDDINGS EXPOSURE
73%
19 of 26

DSA KNOWLEDGE
92%
strong / decent

AI KNOWLEDGE
100%
strong / decent

INDUSTRY CASES
42%
10 of 24 responses

RETRIEVAL / ARCHITECTURES
38%
9 of 24 responses

BUILD / DEPLOY
33%
8 of 24 responses

Source: Canvas pre-flight survey. Programme selections were multi-select; learning-expectation themes were coded from 24 substantive answers.

Pre-Flight Survey · Voices in the Room

OPEN RESPONSES · REPRESENTATIVE ANONYMOUS PHRASES — larger phrases indicate central themes, not frequency

"retrieval and ranking architectures"

"more application based"

"deployment and productionization"

"TikTok / Instagram … real-world case studies"

"YouTube and Spotify"

"modern search alongside recommendation"

"recommendation algorithms … real-world datasets"

"end-to-end recommender … project"

"Two Tower Model"

"LLMs for recommendation … RAG or fine-tuning"

"replicate papers"

"create and deploy MCP server"

"Heavy workload"

"system design … production"

"PyTorch … building and training"

"latest research and progress"

"form our own groups"

"ethics … history and context"

● Systems and concepts ● Building and deployment ● Expectations and concerns

Source: Canvas pre-flight survey, anonymous open responses. Phrases are abbreviated from representative excerpts for slide readability.



Essay 1: Micro Case Study: Designing a Production Recommender Surface

Focus

One approved user-facing recommendation surface.

→

Reason

One or two consequential Week 01–02 choices.

→

Defend

Evidence, limits, and a substantive suit lens.

- ⚠️ (Short) **1,500 words** · Due **Mon, 24 Aug 2026, 23:59 SGT**.
- Use no more than **two external sources**; distinguish platform facts from your reasoned proposal.
- Feature a card-suit lens in the technical design, objective, constraint, or evaluation criterion.
- Include an AI-use declaration; AI may assist inquiry but you need to own the argument.



02 From Neighbours to Features to Latent Structure



LAST WEEK PRE-LECTURE EXERCISE

Look for Hidden Structure

Canvas discussion board · post before Mon, 23:59 SGT

Post a short, concrete response to this question; then reply constructively to one classmate.

> **Name one latent dimension that could explain several ratings—but is not an item metadata label.**

Next week: turn this intuition into latent-factor representations of the user-item matrix.



LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

Latent dimensions students noticed 📶



Three Routes to a Recommendation

Collaborative filtering (CF; last week)

Evidence

User–item interactions

Mechanism

Borrow from similar users or items

Watch

Thin local overlap can leave a

Content-based filtering (CBF)

Evidence

Explicit item, user, or context features

IR mechanism

Encode and match genres, tags, and text

Watch

Feature coverage and encoding

Latent-factor model (LFM)

Evidence

User–item interactions

Mechanism

Learn shared hidden coordinates from many interactions

Watch

A factor is a learned pattern,

Explicit Features Can Be Weighted

Observed or stated fields can enter a recommender score before any latent factors are learned.

User / context fields

`student` · `Singapore` · `vegetarian` `weekday` `evening` · `budget: low`

Supplied, measured, or declared—not

+

Item fields

`cuisine: Thai` · `price: $$` `genre: comedy` · `duration: 90 min`

Observed descriptions of a candidate item.

→

Weighted score

$$\text{score}(u, i, c) = \mathbf{w}^{\top}$$

Vector ***x*** represents an item or a user. Use a similarity metric (Week 01) to define similarities.

Text Is One Possible Feature Source

Text can be an **item field**, **user field**, or **contextual need**.

User / context text

`“animated family film”`
`“a light comedy tonight”`

Profiles, reviews, queries, and session descriptions



Movie item text

`Toy Story (1995)`
`Animation · Children’s · Comedy`

Titles, genres, tags, synopses, and reviews can describe a candidate.



Encoded feature

$\mathbf{x}_{\text{text}} = \text{encode}(\text{tex}$

Encoding maps words to coordinates for comparison or combination.



A Bag of Words Is Sparse and Literal

text / term	animated	family	light	comedy	animation	children
need: animated family film	1	1	0	0	0	0
need: light comedy tonight	0	0	1	1	0	0
item: Toy Story (1995)	0	0	0	1	1	1

- One coordinate per vocabulary term; most coordinates are zero.
- The second need shares `comedy`; the first shares no exact coordinate with the listed movie genres.
- `animated` and `animation`, and `family` and `children`, remain separate until another mechanism relates them.



TF and IDF Signal Different Things

TF: local evidence

$$\text{tf}(t, d) = \text{count of term } t \text{ in document } d$$

Repeated occurrence can strengthen a term's contribution **within one document**.

IDF: collection discrimination

$$\text{idf}(t) = \log \frac{N}{df_t}$$

A term found in almost every document contributes less to distinguishing documents.

$$\text{tf-idf}(t, d) = \text{tf}(t, d) \times \text{idf}(t)$$

Worked TF–IDF: Two Titles, One Interest Profile

User interest: `animated family comedy`

A: *Toy Story* · Animation · Children's · Comedy **B:** *Mrs. Doubtfire* · Family · Comedy

Vocabulary coordinates

term	IDF	U	A	B
`animated`	1.8	1	0	0
`animation`	1.8	0	1	0
`family`	1.4	1	0	1
`children`	1.4	0	1	0
`comedy`	1.2	1	1	1

`animated` ≠ `animation`; `family` ≠ `children`.

Cosine = dot product ÷ (two vector norms)

$$\cos(\mathbf{u}, \mathbf{t}) = \frac{\mathbf{u} \cdot \mathbf{t}}{\|\mathbf{u}\| \, \|\mathbf{t}\|}$$

A · Toy Story — 0.217

dot: 1.2 × 1.2 = 1.44

norms: ‖U‖ = 2.577; ‖A‖ = 2.577

1.44 ÷ (2.577 × 2.577)

B · Mrs. Doubtfire — 0.716

dot: 1.4² + 1.2² = 3.40

norms: ‖U‖ = 2.577; ‖B‖ = 1.844

3.40 ÷ (2.577 × 1.844)

Dense Word Embeddings Learn Neighbourhoods

Training contexts

Words that occur in related contexts provide a learning signal.

→

Dense vectors

Each word receives a short, continuous learned representation.

→

Neighbourhoods

Related usage can place vectors closer together.

- A sparse document–term vector has dimension (): one coordinate per vocabulary term, with most coordinates equal to zero.
  - **Typical scale:** $V =\sim 50,000–100,000$ terms after tokenisation and filtering.
- Dense word embeddings use a much smaller learned dimension $d \ll V$, while retaining continuous coordinates for semantic similarity.
- Their coordinates are learned patterns, **not human-readable facts or ground truth**.

Sparse and Dense Make Different Trade-offs

Question	TF–IDF	Dense word embedding
What is represented?	Exact vocabulary-term evidence	Learned continuous coordinates
Main strength	Transparent lexical matching	Can generalise across related usage
Main limitation	Synonyms remain separate terms	Learned proximity can be opaque or misleading
Typical score	Cosine similarity over sparse vectors	Cosine/dot-product similarity over dense vectors

Bridge: *IR turns* ***text into comparable vectors****. LFM will turn* ***interaction history into comparable user and item vectors****.*

Break — 5 minutes

IR HISTORY · 1958

Keyword search predates the Web

At IBM, **Hans Peter Luhn** demonstrated **Key Word in Context (KWIC)**: an automatic index that rearranged keywords while preserving their textual context.

By the early 1960s, KWIC had become central to hundreds of computerised indexing systems—a useful reminder that retrieval began with finding useful words, not with the Web.

05:00

▶ Start Timer



03 Latent Factor Models



One Vector-Space Idea, Two Evidence Sources

Retrieval	Latent-factor recommendation
Query and document vectors	User and item vectors
Evidence: words in text	Evidence: observed interactions
Score: lexical/semantic match	Score: predicted affinity
Goal: rank relevant documents	Goal: rank plausible items

Analogy, not equivalence: *a latent factor is not a word, a recorded metadata field, or a fact about a person.*

Video: SVD for Movie Recommendations

matrix factorization

$$R = M\Sigma U^T$$

singular value decomposition (svd)

Sundog Education sundog-education.com

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Two Factorisation Lenses: Signed or Additive?

Both methods approximate a matrix with fewer factors, but their **constraints change the meaning** of those factors.

SVD-style lens

$$R \approx U_k \Sigma_k V_k^\top$$

- Factors may be **positive or negative**.
- Natural for centred residuals: above- and below-baseline evidence can offset.
- The singular values in (_k) scale the retained directions.

NMF lens

$$X \approx WH, \qquad W, H \geq 0$$

- Factors are **nonnegative and additive**.
- Natural for counts, intensities, or “parts” interpretations.
- Components combine; they do not cancel one another.

Shared goal: *compress matrix structure.* ***Different question:*** *may a factor subtract,*



Worked SVD Lens: Signed Residual Directions

Toy centred residual matrix — positive means “above this user’s baseline”; negative means “below it.”

[Math Processing Error]

Read the signs

- User 1: Item A above baseline; Item B below.
- User 2: the reverse pattern.
- One signed direction captures the contrast.

Why SVD fits this view

The negative entries are meaningful **residual directions**, not impossible ratings. A nonnegative-only factorisation could not reproduce this signed residual matrix directly.



Worked NMF Lens: Additive Intensity Components

Toy nonnegative interaction matrix — every entry is an intensity, so components are required to add rather than subtract.

[Math Processing Error]

Read the factors

- (W): User 2 has twice the component intensity of User 1.
- (H): Item A expresses twice as much of the component as Item B.
- Their product reconstructs every entry without negative values.

Why NMF fits this view

For counts or accumulated activity, a factor can be interpreted as an **additive amount**. There is no “minus 3 clicks” component to cancel another one.



🎰 Bandit Time: What Does Factorisation Learn?

A restaurant app has a sparse matrix of diner ratings. Which description best matches latent-factor factorisation?

A. It sorts restaurants only by their recorded cuisine label.

B. It learns compact vectors for diners and restaurants from many observed interactions, then uses their alignment to estimate affinity.

C. It treats every unvisited restaurant as a one-star rating.

D. It copies the rating of the geographically nearest diner.

A Dot Product Turns Factors into a Score

$$\hat{r}_{ui} = \mathbf{u}_u^\top \mathbf{v}_i$$

User vector

$$\mathbf{u}_u = [u_{u1}, u_{u2}, \ldots, u_{uk}]$$

A learned location in the model's preference space.

Item vector

$$\mathbf{v}_i = [v_{i1}, v_{i2}, \ldots, v_{ik}]$$

How an item aligns with the learned interaction patterns.

The dot product produces a **model score**. A production recommender can still filter, diversify, or re-rank the resulting candidates.



Worked Calculation: Two Items, One Score

$$\mathbf{u}_u = [0.8, 0.2], \qquad \mathbf{v}_A = [0.9, 0.1], \qquad \mathbf{v}_B = [0.4, 0.8]$$

Item A

$$\hat{r_{uA}} = 0.8(0.9) + 0.2(0.1) = \mathbf{0.74}$$

Item B

$$\hat{r_{uB}} = 0.8(0.4) + 0.2(0.8) = \mathbf{0.48}$$

Current model verdict: Item A scores higher. **Still unresolved:** diversity, constraints, uncertainty, exposure, and the rest of the candidate list.

Training Fits Evidence, Regularisation Limits It

$$\min_{U,V} \sum_{(u,i)\in\Omega} \left(r_{ui} - \mathbf{u}_u^\top \mathbf{v}_i\right)^2 + \lambda \left(\|U\|_F^2 + \|V\|_F^2\right)$$

- **Observed interaction set Ω:** explicit ratings or implicit events select the evidence the objective fits.
- **User factors U and their row vectors:** are adjusted together with the item factors to explain the selected evidence.
- **Item factors V and their row vectors:** align with user factors to produce the model score.
- **Regularisation strength λ:** limits convenient memorisation by penalising excessively large factor values.

Group Discussion · Choose the Lens

In groups of 3–4, assign a lens, state the evidence, and name one caveat. **Time: 5 minutes.**

Case A · Centred ratings

Each value is a deviation from the user's own average rating: some are positive, some negative.
Choose SVD or NMF. Why?

Case B · Watch counts

Entries are nonnegative counts of completed views across topic groups.
Choose SVD or NMF. Why?

Case C · Missing ratings

Most user–item pairs were never observed; they are not zero ratings.
What must both approaches avoid?

05:00

▶ Start 5 min

Report back: one choice, one data property that justifies it, and one interpretation or missing-data caveat.



NEXT WEEK · PREVIEW

Week 03 — Evaluation of Recommendation Systems

An offline metric tells us how a list performed. We ask with our four-suit lenses whether it performs well under a scenario.

♣ Robustness
Can the protocol resist distorted or manipulated evidence?

♥ Social impact
Does the ranking inform without unwanted exposure or conformity?

♦ Discovery
Does the list surface genuinely new, long-tail, or emerging choices?

♠ Constraints & explanation
Can users hold important attributes fixed and understand why an item appears?

